

README: Computational Data and CPLEX Model Reference

Integrated Framework for Rural Passenger and Freight Synergy:
A Multi-Period Location-Inventory-Routing Approach

1 Repository scope

This repository accompanies the manuscript *Integrated Framework for Rural Passenger and Freight Synergy: A Multi-Period Location-Inventory-Routing Approach*. This README provides supplementary information on the released computational data and the CPLEX/OPL implementation used in the computational experiments.

The released IBM ILOG CPLEX OPL source documented below implements the deterministic exact model used as the baseline computational formulation. Its source comments use internal implementation labels (for example, “Constraint (11)”) that are independent of the equation numbering in the manuscript. Table 3 therefore provides an explicit manuscript–code correspondence rather than assuming that the two numbering systems match.

The benchmark data are derived from established vehicle-routing problem (VRP) datasets. Customer locations, depot information, and source demand provide the spatial and demand basis. These data are extended with the features required by the passenger–freight location-inventory-routing problem studied in the paper: co-located candidate service-station representations, fixed scheduled passenger-bus services, multiple service periods, ordinary and special freight, and service-station inventory across periods. The resulting data are problem-specific passenger–freight LIRP instances, not the original VRP benchmarks; their objective values are therefore not directly comparable with published VRP optima.

Instance generation is completed before optimization. The released files contain the realized inputs used in the reported experiments, including the period-specific freight demands. The optimization program reads these values directly and does not redraw stochastic inputs at run time.

2 Benchmark-derived instances

2.1 Naming and reduced-size P-type instances

P- and C-type instances use the labels

$$P-nX-kY-RB \quad \text{and} \quad C-nX-kY-RB,$$

where X is the actual number of customer nodes included in the constructed instance, Y is the number of logistics vehicles, and B is the number of predefined bus routes.

Among the standard P-series benchmarks used in this study, $P-n16-k8$ is the smallest source benchmark. Additional P-type instances with fewer than 16 modeled customer nodes are constructed from $P-n16-k8$. The zero-demand node is first excluded from the modeled customer set; the required number of nonzero-demand customers is then retained in their original order. Other passenger–freight data are generated using the same transformation rules as for the remaining benchmark-derived instances.

2.2 Network and fixed passenger-bus services

The transformed network contains a regional distribution center (RDC), customer nodes, candidate service stations, scheduled passenger-bus services, and logistics vehicles. Four bus-route templates are used in the benchmark transformation (Table 1). The realized fixed route arcs for each released instance are stored in `BusRouteArcs[r]`.

Table 1: Predefined bus-route templates used in the benchmark transformation

Route	Predefined node sequence	Route	Predefined node sequence
1	1–2–6	3	5–7–13–9
2	4–11–3	4	11–15–12–8

These buses are existing scheduled passenger services, not an additional freight-only fleet. Their physical routes, stop order, and service windows are input data. The model decides freight assignment, pickup and delivery quantities, and the freight-load profile along the fixed route; it does not redesign the passenger route.

Candidate service stations are transfer points between logistics vehicles and scheduled buses. For customer locations associated with the predefined bus routes, the customer node and its corresponding candidate service-station node have identical geographic coordinates but distinct node identifiers and operational roles. In the encoded freight-service sequence, the candidate station node is ordered immediately before its co-located customer node, so that replenishment, when selected, is processed before customer delivery at the same physical location. This representation does not create an additional physical stop or travel segment and does not modify the predefined passenger route or stop order. An opened station may receive ordinary freight from a logistics vehicle and hold it for pickup by a scheduled bus in the same or a later period.

2.3 Periods, demand, and coordinates

The service horizon is divided into three periods: morning, midday, and evening. Source demand is allocated across the three periods according to a discrete uniform distribution. Demand in each period is then split into ordinary and special freight following a uniform distribution. The realized values are stored as Demand_n and Demand_s in the released data.

For source datasets given in latitude–longitude coordinates, a planar transformation is applied before the passenger–freight network is built, with the first depot used as the coordinate origin. The transformed coordinates and distance data are part of the computational input.

3 Fields in the released input data

Table 2 lists the main fields read by the current CPLEX OPL model. The same names appear in the OPL source, so each field can be located directly in the model definition. Values in the right-hand column are common benchmark settings where applicable; the empirical case study uses the case-specific values reported in the manuscript.

Table 2: Main OPL input fields and benchmark assignments

Input field	Meaning / assignment
Nodes	Complete node set used by the transformed instance.
Customers	Customer nodes included in the transformed instance. For reduced-size P-type instances, the rule in Section 2.1 is used.
Stations	Candidate service-station nodes described in Section 2.2. For route-associated customer locations, the station representation is co-located with the corresponding customer node (identical coordinates) but uses a distinct node identifier and operational role.
DistributionCenter	Regional distribution center used by the logistics fleet; based on the source depot.
BusOrigins, BusDestinations	Origin and destination node sets used together with BusRouteArcs[r] to identify the endpoints of the predefined fixed bus routes.
BusRoutes	Identifiers of the scheduled passenger-bus services.

Input field	Meaning / assignment
LogisticsVehicles	Available logistics vehicles.
Stages	Number of planning periods; set to 3 in the benchmark experiments. The OPL model also defines $T_{plus}=1..(Stages+1)$ for the customer advance-delivery balance.
LogisticsNodes, NonLogisticsNodes	Node subsets used by the logistics-vehicle routing implementation to delimit feasible visits and flow-balance constraints.
Arcs	Common directed road-arc set used by the logistics-vehicle model.
BusRouteArcs[r]	Fixed directed route-arc set for scheduled bus service r ; it represents the exogenously predefined bus route and is not a route-design decision variable.
Distance	Distance associated with each road arc, taken from the source spatial data or the transformed coordinates.
TransportCost	Unit logistics transportation cost on each arc; $c_{ij} = 1$ in the benchmark experiments.
StationLeaseCost	Period-specific service-station leasing cost; $f_j^t = 10$ in the benchmark experiments.
BusServiceCost	Scheduled-bus freight-service cost. The general manuscript formulation retains customer and bus-service indices; the benchmark experiments use the common value $c_{ir} = 0.5$ for applicable customer-service pairs.
Demand_n, Demand_s	Realized period-specific ordinary- and special-freight demands after the demand-allocation procedure in Section 2.3.
BusCapacity	Freight capacity of the physically separated cargo space on a scheduled passenger bus; $Q_r = 35$ in the benchmark experiments.
VehicleCapacity	Logistics-vehicle capacity; $Q_k = 150$ in the benchmark experiments.
TravelTime	Travel time associated with each road arc; the realized values used by the model are stored in the released input data.
ServiceTime	Node service time; 0.2 at customer nodes and 0.1 at service-station nodes in the benchmark experiments.
MaxWorkingTime	Maximum working time of a logistics vehicle in one period; $T_{max} = 8$ in the benchmark experiments.

4 Manuscript–CPLEX model mapping

Table 3 gives searchable locators for the principal deterministic formulation blocks. The third column reproduces the exact English comment headings in the supplied OPL source. The internal CPLEX labels are intentionally retained because they allow a reviewer to search the released source directly.

The notation used most often maps as follows: A is represented by `Arcs`; A_r by `BusRouteArcs[r]`; h_j^t by `h[j][t]`; q_{jkt}^n by `q_n[k][j][t]`; b_{jrt}^n by `b_n[r][j][t]`; β_{irt}^n by `BusServeCustomer[r][i][t]`; and $\delta_{ikt}^n, \delta_{ikt}^s$ by `VehicleServeCustomer_n[k][i][t]` and `VehicleServeCustomer_s[k][i][t]`, respectively.

Table 3: Current manuscript components and searchable CPLEX code locators

Model component	Manuscript location	CPLEX code locator (exact comment heading)
Objective function	Eq. (1)	// Objective function, corresponding to manuscript Eq. (1)
Scheduled-bus initial load, arc load, capacity, and terminal load	Eqs. (2), (12)–(13)	// Constraint (8): Initial freight load of scheduled buses

Model component	Manuscript location	CPLEX code locator (exact comment heading)
Customer delivery and bus-load balance	Eqs. (3)–(4)	// Constraint (9): Freight load on the first bus arc // Constraint (10): Bus capacity on each bus-route arc
		// Constraint (13): Bus load at destination is zero // Constraint (11): Bus load balance at customer nodes
Station pickup, in-bound/outbound load, and pickup capacity	Eqs. (5)–(11)	// Constraint (15): Bus delivery can occur only on fixed bus routes // Constraint (12): Bus load balance at service stations
		// Constraint (14): Bus capacity after station pickup // Constraint (16): Bus pickup can occur only at opened service stations on the fixed route // Constraint (17): Bus pickup and delivery records // Constraint (18): Bus load record at each node
Logistics-vehicle capacity	Eq. (14)	// Constraint (25): Logistics vehicle load initialization
Customer service/quantity requires a logistics-vehicle visit	Eqs. (15)–(16)	// Constraint (26): Logistics vehicle total capacity // Constraint (22): Logistics vehicle delivery requires visiting the customer
Station opening, visit, replenishment, and at-most-one logistics vehicle	Eqs. (17)–(20)	// Constraint (2): Non-station nodes cannot be opened as service stations // Constraint (3): Station-related quantities can only be defined on station nodes // Constraint (23): Logistics vehicle replenishment requires visiting an opened service station // Constraint (24): Each service station is replenished by at most one logistics vehicle in each period
Fixed scheduled-bus route representation (not bus-route design)	Eqs. (21)–(22); notation A_r, o_r, d_r	// Constraint (4): Bus-route variables are restricted to fixed bus-route arcs // Constraint (5): Fixed bus route starts from its origin // Constraint (6): Fixed bus route ends at its destination // Constraint (7): Flow conservation on fixed bus routes
Logistics-vehicle route flow and depot balance	Eqs. (23)–(24)	// Constraint (19): Logistics vehicle route flow balance // Constraint (20): Logistics vehicle departs from and returns to the distribution center
Cross-period service-station inventory	Eqs. (25)–(27)	// Constraint (27): Service-station inter-period inventory balance
Delivery/service linkage	Eqs. (28)–(30)	// Constraint (28a)–(28c): Delivery quantity and service-decision consistency
Single-service and same-vehicle rules	Eqs. (31)–(34)	// Constraint (29): Ordinary-goods delivery is assigned to at most one service mode per period // Constraint (30): Special-goods delivery is assigned to at most one logistics vehicle per period // Constraint (31a): Bus delivery and special-goods logistics delivery cannot occur simultaneously // Constraint (31b): If ordinary and special goods are both delivered by logistics vehicles, // they must be delivered by the same logistics vehicle

Model component	Manuscript location	CPLEX code locator (exact comment heading)
Customer advance-delivery balance and no-backlogging initialization	Eqs. (35)–(38)	// Constraint (32): Customer advance-delivery balance for ordinary goods // Constraint (33): Customer advance-delivery balance for special goods // Constraint (34): Nonnegative customer advance-delivery surplus // Constraint (35): Initial customer advance-delivery surplus
Logistics-vehicle working time	Eq. (39)	// Constraint (36): Logistics vehicle working-time limit
Variable domains	Eqs. (40)–(42)	// Constraint (39): Variable domains
Additional route-feasibility constraints used in the CPLEX implementation	CPLEX implementation	// Constraint (37): MTZ subtour elimination for logistics vehicles // Constraint (38): Depot activation for logistics vehicles

5 Code-mapping note

For ease of verification, Table 3 provides the correspondence between the principal deterministic model components in the manuscript and their implementation in the CPLEX/OPL model. Because the OPL implementation uses auxiliary variables and implementation-oriented constraint groupings, the correspondence is based on modeling logic rather than a strict one-equation-to-one-statement mapping. The searchable comment headings in the OPL file are therefore provided as locators for the relevant implementation blocks.

6 Solver and execution notes

The supplied exact model is written for IBM ILOG CPLEX OPL 22.1.2.0. The model reads the released realized data, generates the OPL model, and applies a CPLEX time limit of 10,800 seconds through `cplex.tilim = 10800`. Binary domains are declared using `dvar boolean`; nonnegative continuous variables are declared using `dvar float+`.

Because instance generation is performed before optimization, reproducing a released exact-model run requires using the corresponding released input data with the supplied OPL model. The optimization stage itself does not regenerate demand, candidate service-station node representations, or bus-route templates.